Stroke

ORIGINAL CONTRIBUTION

Clonal Hematopoiesis and Risk of Stroke: Evidence From Over 800 000 Individuals Across 3 Cohorts

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BACKGROUND: Clonal hematopoiesis of indeterminate potential (CHIP) is a common age-related condition that increases risk for cardiovascular disease. However, its relationship with stroke remains uncertain.

METHODS: To resolve these conflicting findings, we analyzed genomic and clinical data from 800 160 participants with genetic sequencing and medical records across 3 large-scale cohorts: the Vanderbilt BioVU biobank (United States; enrollment 2007–2022), the National Institutes of Health All of Us Research Program (United States; enrollment 2018–2023), and the UK Biobank (United Kingdom; enrollment 2006–2010). The median follow-up time was 4.4 years (interquartile range, 1.8–10.2) in BioVU, 1.9 years (interquartile range, 0.4–3.9) in All of Us, and 12.4 years (interquartile range, 11.7–13.1) in UK Biobank. Stroke events were identified and classified as ischemic or hemorrhagic using *International Classification of Diseases* codes. Subgroup analyses were conducted by driver gene, clone size, sex, and menopausal status. Genetically predicted levels of 27 cytokines were assessed for modification of CHIP-associated stroke risk.

RESULTS: CHIP was associated with increased risk of incident stroke in the meta-analysis (hazard ratio [HR], 1.20 [95% CI, 1.14–1.27]; $P=1.92\times10^{-10}$). This association was observed for ischemic (HR, 1.18) and hemorrhagic (HR, 1.25) stroke subtypes. Gene-specific analyses showed strong associations for *JAK2* (HR, 2.52) and *TET2* (HR, 1.23). *DNMT3A* demonstrated weak but significant associations (HR, 1.13). CHIP was associated with stroke risk in both sexes; however, among women, the association was evident in postmenopausal (HR, 1.49 [95% CI, 1.16–1.92]; $P=1.91\times10^{-3}$) but not in premenopausal participants (HR, 0.70 [95% CI, 0.36–1.43]; $P=0.33$). Among participants with CHIP, but not among participants without CHIP, genetically predicted levels of IL-1RAP (interleukin-1 receptor accessory protein) were predictive of risk for stroke, suggesting IL-1RAP as a modifier of the CHIP-associated risk for stroke.

CONCLUSIONS: Our large-scale, multicohort study establishes CHIP as a determinant of incident stroke risk and IL (interleukin)-1–mediated inflammation as a targetable pathway to reduce this risk.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: clonal hematopoiesis ■ interleukin-1 ■ risk ■ stroke ■ UK Biobank

Stroke is one of the most devastating noncommunicable diseases, ranking as the second leading cause of death worldwide (≈7 million annually) and the third leading cause of death and disability combined, accounting for >160 million disability-adjusted life years.[1–3] Stroke occurs more frequently in men and the elderly, although women face disproportionately higher risks of stroke-related mortality and long-term disability.[4–6] Ischemic and hemorrhagic strokes represent distinct entities, with divergent pathophysiological mechanisms and risk factors.[3]

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Nonstandard Abbreviations and Acronyms

AoU	National Institutes of Health All of Us Research Program
CHIP	clonal hematopoiesis of indeterminate potential
gIL-1RAP	genetically predicted IL-1 receptor accessory protein
HR	hazard ratio
ICD	*International Classification of Diseases*
IL	interleukin
IL-1RAP	interleukin-1 receptor accessory protein
UKB	UK Biobank
VAF	variant allele fraction

While ischemic stroke is primarily driven by atherosclerosis and thrombosis, hemorrhagic stroke is more strongly linked to hypertension, vascular fragility, and coagulation abnormalities.[7] Inflammation plays a central role in the pathogenesis and outcomes of both types of stroke.[8,9] Cytokines represent key mediators in this process.[10–12] Proinflammatory cytokines, such as IL (interleukin)-1[13–15] and IL-6,[16–18] have been shown to exacerbate ischemic brain injury by activating immune cells, disrupting the blood–brain barrier, and amplifying secondary tissue damage. Conversely, anti-inflammatory cytokines like IL-10 may mitigate adverse inflammatory responses and improve stroke prognosis.[19,20] Although stroke is influenced by multiple risk factors, including lifestyle, vascular comorbidities, and menopause, genetic susceptibility has also been firmly established as an important determinant of overall stroke risk.[21–23] Beyond inherited germline variation, emerging evidence suggests that acquired somatic mutations may also contribute to stroke susceptibility.[24,25]

Clonal hematopoiesis of indeterminate potential (CHIP) is the presence of an expanded somatic blood cell clone in individuals without other hematologic abnormalities, most commonly involving mutations in *DNMT3A* and *TET2*.[26,27] CHIP is observed in >10% of individuals aged ≥70 years and is strongly associated with atherosclerotic cardiovascular disease.[28–31] The association between CHIP and stroke was first reported by Jaiswal et al,[24] who demonstrated a 2.6-fold increased risk of ischemic stroke among CHIP carriers in a cohort of 3913 individuals. Subsequently, in a meta-analysis of >80 000 participants across 8 prospective studies, Bhattacharya et al[25] reported a statistically significant increase in total stroke risk (hazard ratio [HR], 1.14, [95% CI, 1.03–1.27]; *P*=0.01); the risk associated with CHIP was strongest for hemorrhagic and small-vessel ischemic stroke subtypes. More recent large-scale analyses of the UK Biobank have yielded conflicting results. For example, Kessler et al[32] reported that only *JAK2*-CHIP was associated with the prevalence of overall stroke and ischemic stroke, whereas Kar et al[33] found that CHIP was not significantly associated with overall stroke incidence (HR range 0.90–1.72; *P*>0.09). A recent study further suggests that CHIP is associated with increased white matter lesion burden, and a proinflammatory profile in patients with ischemic stroke, and that mutations, such as *TET2*, confer higher risks of recurrent vascular events and mortality after ischemic stroke.[34] Recent multiomics and Mendelian randomization studies have suggested causal links between CHIP and ischemic stroke via inflammation.[35]

These divergent findings underscore the need to clarify the relationship between CHIP and stroke at-scale. To address this gap, we analyzed CHIP and stroke outcomes in more than 800 000 individuals from 3 large independent cohorts. We aimed to (1) systematically evaluate the association between CHIP and incident stroke, in aggregate and by major subtype (ie, ischemic and hemorrhagic), (2) explore whether the relationship varies by mutated gene, clone size, sex, or menopausal status, and (3) investigate potential targetable mechanisms via genetically predicted cytokines.

METHODS

Data Availability

Individual-level sequence data, CHIP calls, and polygenic scores have been deposited with UK Biobank and are available to approved researchers through application, as done with other genetic data sets to date under data fields (30 105, 30 106, 30 107). The genotypes and phenotypes of UKB (UK Biobank) and AoU (National Institutes of Health All of Us Research Program) participants are available by application to the UKB (https://www.ukbiobank.ac.uk/register-apply/) and AoU (https://allofus.nih.gov/), respectively. For Vanderbilt BioVU, CHIP sequencing calls are available through controlled access to qualified researchers. Due to Vanderbilt BioVU cohort restrictions, access will require a data use agreement with Vanderbilt University Medical Center, which can be facilitated by the corresponding author.

Study Population and Design

This study utilized data from 3 large-scale cohorts: the Vanderbilt BioVU biobank (United States; enrollment 2007–2022), the AoU (United States; enrollment 2018–2023), and the UKB (United Kingdom; enrollment 2006–2010). Participants with available whole-genome or exome sequencing data were initially considered (BioVU: 245 540; AoU: 275 679; UKB: 488 175). We excluded individuals with a documented history of stroke before enrollment, a history of hematologic malignancy or other non-neoplastic clonal disorders (*International Classification of Diseases* [*ICD*] codes outlined in Table S1), lack of follow-up information, or missing key covariates. As CHIP is highly uncommon in younger individuals, to minimize potential technical confounding, we excluded participants with a baseline age <40 years. Our final analytic cohort was 800 160 participants across BioVU, AoU, and UKB (Figure 1). Although UKB results have been reported in prior publications,[25,32,33]

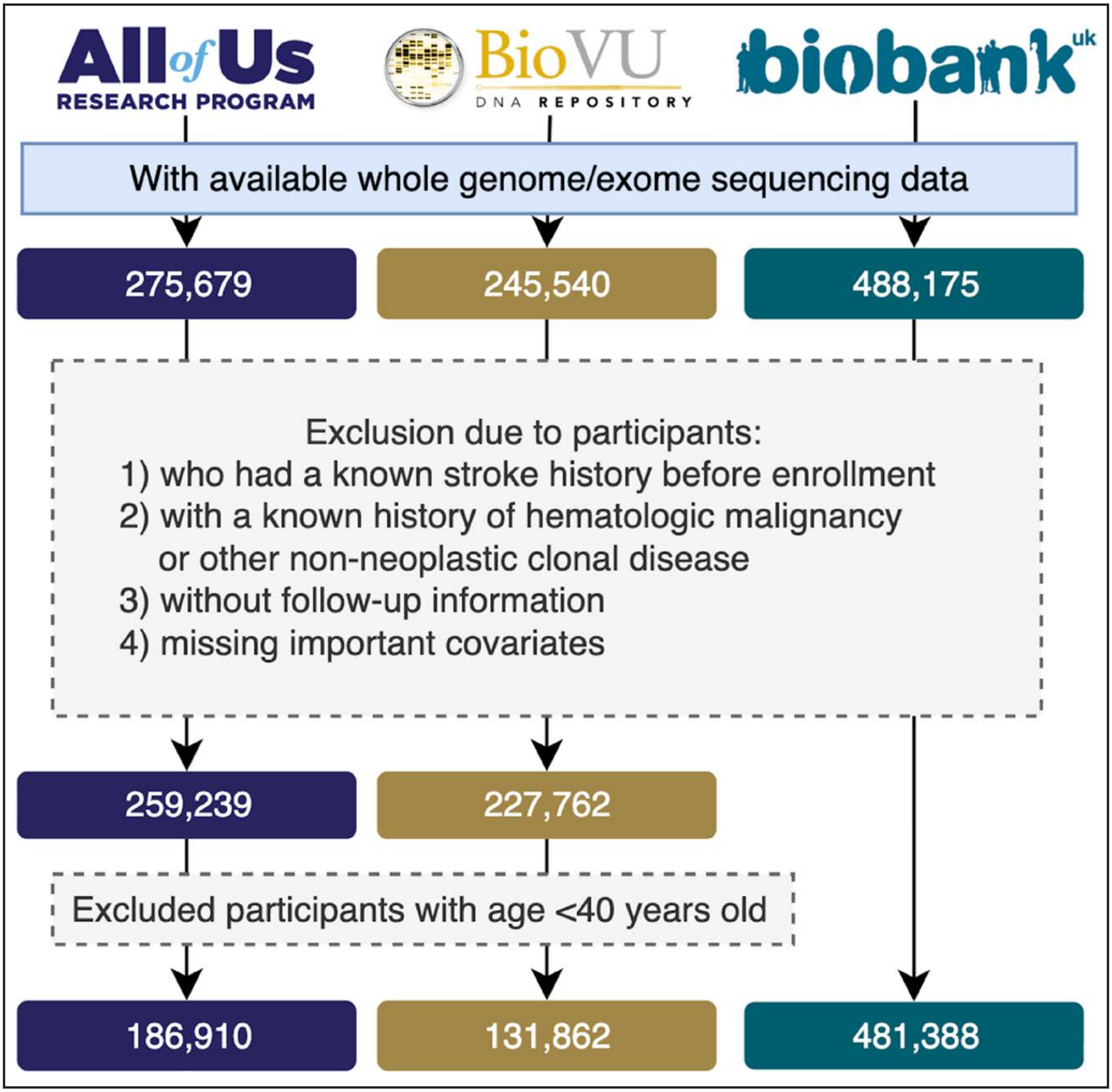

Figure 1. Study population flowchart across 3 cohorts.
Flow diagram showing participant selection in the All of *Us* Research Program, Vanderbilt BioVU, and UK Biobank cohorts. Individuals were excluded if they had (1) a known history of stroke before enrollment, (2) a history of hematologic malignancy or other non-neoplastic clonal disease, (3) missing follow-up information, or (4) missing key covariates. Participants aged <40 years were further excluded. The final analytic cohorts included 186 910 participants from All of Us, 131 862 from BioVU, and 481 388 from UK Biobank, for a total of 800 160 participants.

our analyses leveraged updated genomic data, which included several hundred thousand additional participants not previously examined, as well as updated outcomes data, including 18 553 additional stroke events since the last analysis (N=4321).[33]

CHIP Variant Calls

Putative somatic SNVs and short indels were called with GATK *Mutect2* and filtered according to previously established criteria among the 3 cohorts, as we have previously described.[36,37] Briefly, 74 canonical CHIP genes were screened for potential CHIP mutations using the *Mutect2* somatic variant caller (Table S2).[28] Variants included in the preliminary data set met the following criteria: presence in a preestablished list of candidate CHIP variants, total sequencing depth ≥20, alternate allele read depth count ≥5, and representation in both sequencing directions (ie, F1R2≥1 and F2R1≥1). CHIP mutations were defined as those with a variant allele fraction (VAF) ≥0.02. CHIP detection across all cohorts was derived from whole-genome/exome sequencing data, using a consistent detection method and the same canonical CHIP driver genes list, which specifies candidate missense and indel variants for each gene, as well as a set of genes in which truncating and splice site variants may be considered.

Study Outcome and Covariates

The primary outcome was incident stroke, with additional analyses performed for ischemic and hemorrhagic stroke subtypes. In BioVU and AoU, stroke outcomes were ascertained from linked electronic health records using *ICD* (*ICD-Ninth Revision* and *ICD-Tenth Revision*) diagnosis codes derived from inpatient hospitalizations and death records. In the UK Biobank, stroke events were identified using *ICD-Ninth Revision* and *ICD-Tenth Revision* codes from hospital admissions data and national death registries. Outcome definitions were harmonized across cohorts to ensure consistent classification and comparability with prior studies[25] (Table S3). Covariates included baseline age, sex, smoking status, principal components of ancestry (1–5), low-density lipoprotein cholesterol, hypertension, body mass index, baseline type 2 diabetes, and baseline atrial fibrillation (defined by *ICD-Ninth Revision*: 427.3 or *ICD-Tenth Revision*: I48). Smoking status was categorized as a current smoker versus nonsmoker. Body mass index (kg/m^2) was calculated from measured height and weight at the baseline visit. Hypertension was defined as a diagnosis based on *ICD* codes before baseline or elevated blood pressure at baseline (systolic or diastolic in mm Hg). Type 2 diabetes was defined as a diagnosis based on *ICD* codes before baseline or an elevated baseline fasting glucose (mg/dL). Low-density lipoprotein cholesterol (mmol/L) levels were obtained from baseline laboratory assessments.

Statistical Analysis

Descriptive statistics were used to summarize baseline characteristics. Continuous variables were reported as mean (SD), and categorical variables as counts and percentages. The association between CHIP and incident stroke was assessed using age-scaled Cox proportional hazards models, adjusting for the covariates described above. The proportional hazards assumption was evaluated using Schoenfeld residual–based diagnostics. Left truncation was accounted for by defining each participant's entry age as age at baseline and exit age as age at the time of stroke or censoring. Participants who died before experiencing a stroke were right-censored at the time of death, consistent with a cause-specific Cox proportional hazards modeling framework in the presence of competing risks. The outcome was defined as the first recorded stroke event. Participants were followed from baseline until the earliest occurrence of stroke or the end of follow-up. For subtype-specific analyses, only the first ischemic or hemorrhagic stroke was considered an event of interest. Inverse variance–weighted meta-analysis was performed under either a fixed- or random-effects framework, selected according to the degree of heterogeneity (I^2 statistic).

We conducted multiple sensitivity analyses to assess the robustness of our findings, including (1) additional adjustment for chronic kidney disease, (2) exclusion of individuals diagnosed with hematologic malignancy within 1 year after baseline, and (3) exclusion of individuals with persistent cytosis or cytopenia within 1 year of blood draw, to minimize potential bias from occult clonal or hematologic disease. Cytosis was defined by using a modified version of the World Health Organization criteria (erythrocytosis: hemoglobin >16.5 g/dL (females) or 18.5 g/dL (males); thrombocytosis: platelets >450 000 cells/mL; and leukocytosis: white blood cell count >11 000 cells/mL). Cytopenias were defined by using a modified version of World Health Organization criteria 8 (anemia: hemoglobin <12.0 g/dL [females] or 13.0 g/dL [males]; thrombocytopenia: platelets <150 000 cells/mL; and leukopenia: white blood cell count <3700 cells/mL).

Additional analyses were conducted stratified by VAF of CHIP mutations (≥2% and ≥10% versus no mutation), sex, and menopausal status for women (menopausal data were only available in UKB). Menopausal status was derived from self-reported data, encoded using UK Biobank Data-Coding 100 579 (Field ID: 2724). To investigate whether menopausal status altered the association between CHIP and stroke in women, we conducted a sensitivity analysis using age-matched groups. Specifically, we applied 1:1 nearest-neighbor matching on baseline age between postmenopausal and premenopausal women using the *MatchIt* package in R. A caliper of 1.5 years was applied to ensure close age matching. The matched data set included equal numbers of postmenopausal and premenopausal women (n=97 076 per group), and subsequent analyses of the CHIP-stroke association were performed within this matched population, further adjusting for baseline age and other covariates to minimize any residual confounding. This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Genetically Predicted Cytokine Modifier Analysis

The genetically predicted cytokines were estimated among participants of European ancestry in the BioVU (N=107 479), AoU (N=118 724), and UKB (N=458 681). To derive genetically predicted cytokines, we first identified representative SNPs for each susceptibility locus based on a Bayesian ridge model trained with SomaLogic protein measurements from the INTERVAL study (OPGS000019).[38] The genetically predicted cytokines were then calculated by summing the weighted genotypes of all selected variants using the following formula:

$$\text{Genetically predicted cytokine } X = \sum_{i=1}^{n} \beta i \; SNPi$$

where βi represents the estimated weight (ie, the natural logarithm of the odds ratio) of the i-th SNP, derived from the reference data sets, and *SNPi* is the genotype dose of each risk allele for that SNP. In total, 27 distinct genetically predicted cytokines were computed using this approach (SNPs and weights listed in Table S4).

For genetically predicted cytokines, analyses were performed using logistic regression testing the association between genetically predicted cytokine level and whether the participant had a stroke event, adjusting for covariates listed above. The analysis was performed separately for CHIP carriers and non-CHIP carriers. We tested the statistical interaction between CHIP status and genetically predicted cytokines on stroke risk by including an interaction term in the regression model. The odds of stroke per 1 SD change in cytokine levels were estimated separately in CHIP carriers and noncarriers, and interaction effects were assessed using the ratio of odds ratios and corresponding 95% CIs. Multiple testing correction was applied with a Bonferroni threshold of P<0.05/27. All analyses were conducted by R version 4.2.0 (https://www.r-project.org).

RESULTS

After applying the exclusion criteria, a total of 800 160 participants were included across 3 cohorts: 131 862 from BioVU, 186 910 from AoU, and 481 388 from the UKB. The mean age at blood draw was 59.2±11.4 years in BioVU, 60.7±11.3 years in AoU, and 56.5±8.1 years in the UKB. Females accounted for 54% to 60% of participants across cohorts. Most participants were of European ancestry, although the AoU and BioVU cohorts included relatively high proportions of African ancestry individuals (19.3% and 10.4%, respectively). The median follow-up time was ≈4.4 years (interquartile range 1.8–10.2) in BioVU, 1.9 years (interquartile range 0.4–3.9) in AoU, and 12.4 years (interquartile range 11.7–13.1) in the UKB. During follow-up, an incident stroke occurred in 7.3% of BioVU (n=9351), 2.7% of AoU (n=5054), and 0.8% of the UKB (n=3862; Table). Expectedly, BioVU, derived from electronic health records of a tertiary care health system, had the highest incidence of stroke.

Baseline Clonal Hematopoiesis Characteristics

Across the 3 cohorts, a total of 35 270 (4.4%) participants had ≥1 detectable CHIP mutation(s), with rates of 7.5% in BioVU, 5.2% in AoU, and 3.2% in the UKB. CHIP carriers were, on average, older than noncarriers (P<0.001), and the prevalence was higher for males than females (P<0.001). The mutational landscape of CHIP was dominated by *DNMT3A*, *TET2*, and *ASXL1*, which together accounted for ≈82% of all detected mutations (Figure S1). CHIP prevalence was associated with age in each cohort. Age-related increases in CHIP prevalence were consistent across cohorts, although the absolute frequencies differed, with BioVU showing the highest prevalence at all age groups. When stratified by sex, both men and women demonstrated similar age-related trends, with only modest differences in overall prevalence after 80 years (Figure S2).

CHIP Associated With Increased Risk of Incident Stroke

In our primary analyses of the 3 cohorts, carriers of CHIP mutations had a significantly higher risk of incident stroke compared with noncarriers (HRs ranging from 1.15 to 1.36 across cohorts). Meta-analysis demonstrated a robust association between CHIP and stroke risk (HR, 1.20 [95% CI, 1.14–1.27]; P=1.92×10^{-10}; Figure 2; Table S5). We further analyzed traditional stroke risk factors within the cohorts and found that the risk associated with CHIP was comparable to that conferred by male sex (male versus female, HR, 1.16 [95% CI, 1.12–1.20]; Figure S3).

Risk of stroke differed by CHIP driver mutation. *TET2* CHIP was associated with a relatively high-risk (HR, 1.23 [95% CI, 1.1–1.37]; P=2.75×10^{-4}), with consistent effects across the 3 cohorts. *JAK2* CHIP conferred the highest increased risk (HR, 2.52 [95% CI, 1.85–3.43]; P=5.54×10^{-9}), despite the relatively small number of carriers (N=464). *DNMT3A* CHIP showed a weaker but statistically significant association (HR, 1.13 [95% CI, 1.04–1.22]; P=3.05×10^{-3}), whereas *ASXL1* CHIP was not significantly associated with stroke (HR, 1.07 [95% CI, 0.9–1.26]; P=0.43). Kaplan-Meier analyses based on the BioVU, AoU, and UKB cohorts illustrate these findings, showing a lower stroke-free survival probability among CHIP carriers compared with noncarriers (log-rank P<0.0001). The decline was particularly pronounced in individuals harboring *TET2* mutations, whose risk trajectory was substantially steeper than that of non-CHIP carriers and carriers of *DNMT3A* mutations (Figure 3A).

All sensitivity analyses yielded materially similar results. The association between CHIP and incident stroke remained statistically significant after additional adjustment for chronic kidney disease (Table S6), after exclusion of individuals diagnosed with a hematologic malignancy within 1 year following baseline (Table S7), and after exclusion of individuals with abnormal blood counts at blood draw (Table S8), with effect estimates remaining essentially unchanged (HR ≈1.20). We further examined whether the association between CHIP and stroke risk differed by clone size. Interestingly, the results were consistent across thresholds: the hazard of CHIP-associated incident stroke remained essentially unchanged when restricting to clones with VAF ≥10% (HR, 1.22 [95% CI, 1.15–1.3]) compared with the standard VAF ≥2% definition (HR, 1.20 [95% CI, 1.14–1.27]). Similar patterns were observed for individual driver genes, indicating that the increased stroke risk associated with CHIP was not materially modified by clone size (Figure S4).

Different CHIP Driver Mutations Associate With Ischemic and Hemorrhagic Stroke

To determine whether the impact of CHIP varies across stroke subtypes, we analyzed its associations with ischemic and hemorrhagic stroke separately. When stratified by stroke subtype, CHIP was associated with increased risk of both ischemic and hemorrhagic stroke. The relative risk was slightly higher for hemorrhagic stroke (HR, 1.25 [95% CI, 1.1–1.43]; P=7.47×10^{-4}; Figure 2; Table S9) than for ischemic stroke (HR, 1.18 [95% CI, 1.01–1.38]; P=3.33×10^{-2}; Figure 2; Table S10). In comparative analyses with established risk factors, the effect of CHIP on ischemic stroke risk was of similar magnitude to that observed for male sex (HR, 1.17), while for hemorrhagic stroke, the impact of CHIP approached that of smoking status (HR, 1.24; Figure S3).

As with overall stroke, different drivers exhibited distinct associations. *TET2* CHIP was strongly associated

Table. Basic Socio-Demographic Characteristics by Cohort

	BioVU		All of Us		UK Biobank	
	Total	CHIP carriers	Total	CHIP carriers	Total	CHIP carriers
	No. 131 862	No. 9951 (7.5%)	No. 186 910	No. 9760 (5.2%)	No. 481 388	No. 15 559 (3.2%)
Demographic						
Age of blood draw, y	59.2±11.4	66.5±11.7	60.7±11.3	67.8±10.6	56.5±8.1	60.7±6.7
Sex						
Female	69 432 (54.0%)	5199 (52.2%)	112 117 (60.0%)	5470 (56.0%)	260 771 (54.2%)	8232 (52.9%)
Male	59 205 (46.0%)	4752 (47.8%)	74 793 (40.0%)	4290 (44.0%)	220 617 (45.8%)	7327 (47.1%)
Ancestry						
European	107 479 (83.6%)	8544 (85.9%)	118 724 (63.5%)	7099 (72.7%)	458 681 (95.3%)	15 033 (96.6%)
African	13 374 (10.4%)	879 (8.8%)	36 057 (19.3%)	1569 (16.1%)	9215 (1.9%)	204 (1.3%)
East Asian	1682 (1.3%)	97 (1.0%)	2853 (1.5%)	112 (1.1%)	13 492 (2.8%)	322 (2.1%)
Current smoker						
No	82 199 (63.9%)	6209 (62.4%)	102 413 (54.8%)	4964 (50.9%)	262 613 (54.6%)	7397 (47.7%)
Yes	46 438 (36.1%)	2742 (37.6%)	80 471 (43.1%)	4594 (47.1%)	216 317 (45.0%)	8060 (51.9%)
Baseline health status						
Hypertension	19 223 (14.9%)	1755 (17.6%)	13 609 (7.3%)	979 (9.3%)	48 536 (10.1%)	2160 (13.9%)
Type 2 diabetes	26 745 (20.8%)	2264 (22.8%)	24 444 (13.1%)	1330 (13.6%)	14 203 (3.0%)	604 (3.9%)
Atrial fibrillation	16 413 (12.4%)	1901 (19.1%)	14 558 (7.8%)	1154 (11.8%)	7752 (1.6%)	341 (2.2%)
LDL cholesterol, mmol/L	2.7±0.6	2.6±0.6	2.7±0.6	2.7±0.7	3.6±0.9	3.5±0.9
Body mass index	28.0±7.9	28.5±6.9	29.2±6.8	29.7±7.5	27.4±4.8	27.5±4.6
Health outcome						
Stroke incidence						
Overall	9351 (7.3%)	883 (8.9%)	3751 (2.7%)	336 (3.4%)	3862 (0.8%)	194 (1.2%)
Ischemic stroke	4215 (3.3%)	496 (5.0%)	3338 (1.8%)	220 (2.3%)	1950 (0.4%)	99 (0.6%)
Hemorrhagic stroke	2077 (1.6%)	187 (1.9%)	634 (0.3%)	47 (0.5%)	380 (0.1%)	21 (0.1%)

CHIP indicates clonal hematopoiesis of indeterminate potential; and LDL, low-density lipoprotein.

with both ischemic stroke (HR, 1.19 [95% CI, 1.03–1.36]; $P=1.55\times10^{-2}$) but showed no significant association with hemorrhagic stroke (HR, 1.07 [95% CI, 0.82–1.4]; $P=0.61$). *DNMT3A*- and *ASXL1*-CHIP were not significantly related to either subtype. In contrast, *JAK2* CHIP conferred elevated risks for ischemic (HR, 2.18 [95% CI, 1.46–3.26]; $P=1.33\times10^{-4}$) and hemorrhagic stroke (HR, 2.08 [95% CI, 0.99–4.37]; $P=0.05$), with the latter not reaching statistical significance due to limited *JAK2* CHIP carriers. Across all sensitivity analyses, the association between CHIP and ischemic stroke remained materially unchanged (Tables S11 through S13). For hemorrhagic stroke, the primary association with CHIP was similarly stable, and the association between *JAK2* CHIP and hemorrhagic stroke reached statistical significance in these analyses ($P<0.05$; Tables S14 through S16). We further evaluated whether the association of CHIP with ischemic and hemorrhagic stroke differed by VAF. Consistent with the findings for overall stroke, the risk estimates were similar when restricting to large clones with VAF ≥10% compared with the standard VAF ≥2% definition, with no evidence of heterogeneity across subtypes and driver genes (Figures S5 and S6).

Effect of Menopausal Status on the Association Between CHIP and Stroke

To assess sex-specific effects, we stratified the analyses by sex within each cohort. Across BioVU, AoU, and UKB, CHIP carriers of both sexes had a significantly higher risk of incident stroke than noncarriers. In the meta-analysis, the association of CHIP and stroke incidence was significant in men (HR, 1.26 [95% CI, 1.15–1.39]; $P=2.10\times10^{-6}$) and in women (HR, 1.17 [95% CI, 1.08–1.27]; $P=1.11\times10^{-4}$), with a numerically higher hazard in men (Figure 4A). To examine whether menopausal status modifies the association between CHIP and stroke, we stratified female participants in the UKB by baseline menopausal status. Among premenopausal women, CHIP was not significantly associated with incident stroke (HR, 0.70 [95% CI, 0.36–1.43]; $P=0.33$), whereas the association was significant among postmenopausal women (HR, 1.49 [95% CI, 1.16–1.92]; $P=1.91\times10^{-3}$). To minimize potential confounding by age, we conducted a 1:1 nearest-neighbor matching with baseline age between the 2 groups. The results remained consistent: CHIP was significantly associated with higher stroke risk in postmenopausal women (HR, 1.46 [95% CI,

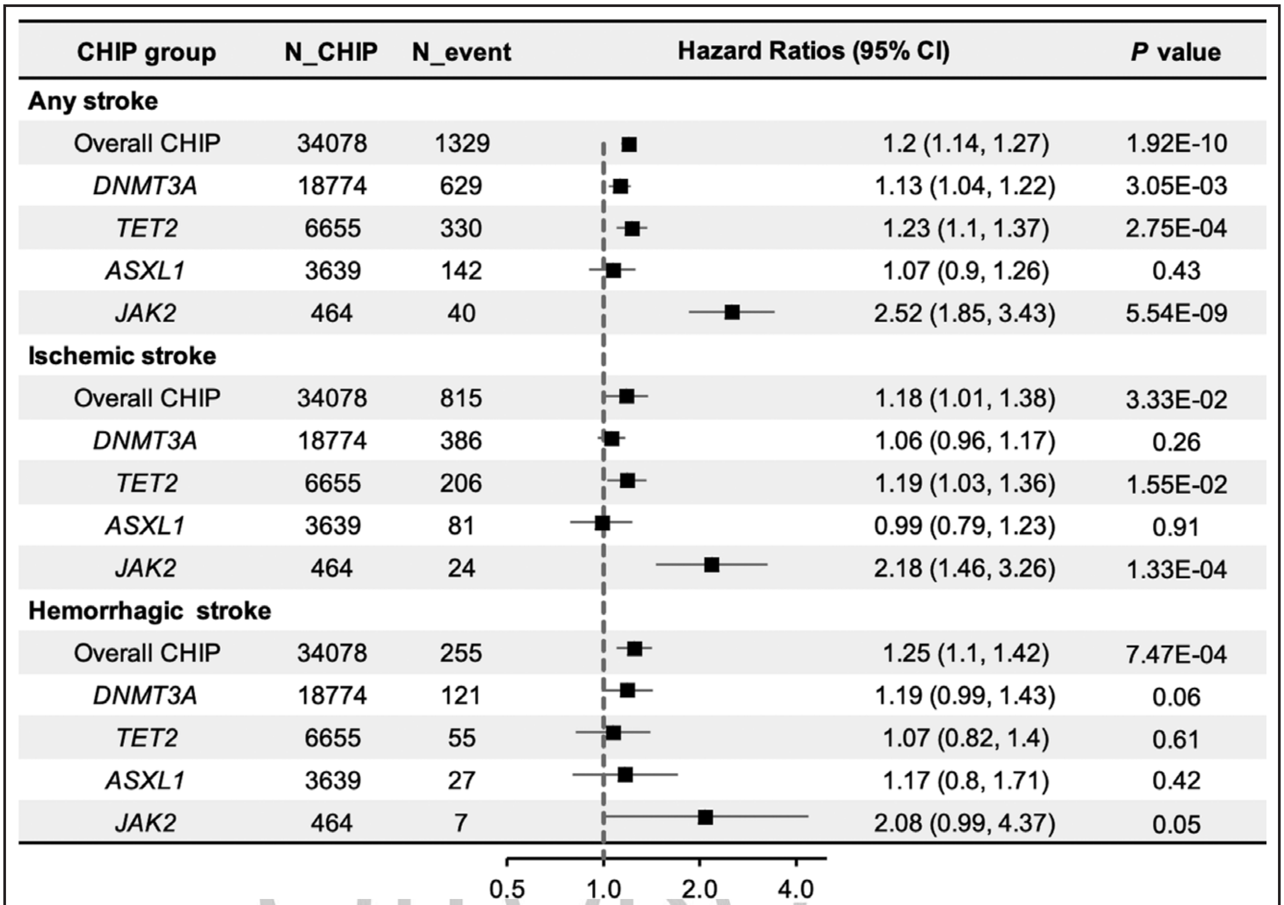

Figure 2. Time-to-event association of clonal hematopoiesis of indeterminate potential (CHIP) with incident stroke after meta-analysis.

This forest plot shows the associations of overall CHIP and individual driver gene mutations (*DNMT3A*, *TET2*, *ASXL1*, *JAK2*) with stroke outcomes. The column N_CHIP indicates the number of CHIP carriers included, and N_event represents the number of stroke cases among these carriers. Hazard ratios with 95% CIs quantify the effect sizes. Results are derived from meta-analysis across the All of Us, BioVU, and UK Biobank cohorts.

1.03–2.08]; *P*=0.033), but not in premenopausal women (*P*=0.34; Figure 4B).

Genetically Predicted Cytokines and Stroke Risk Among CHIP Carriers

Finally, we investigated the role of genetically predicted circulating levels of 27 cytokines in mediating the association between CHIP and stroke. Among the proteins evaluated, gIL-1RAP (genetically predicted IL-1 receptor accessory protein) was significantly associated with increased stroke risk (odds ratio, 1.17 [95% CI, 1.06–1.28]; P=1.50×10^{-3}; Figure 5), whereas no significant associations were observed for other cytokines (Figure S7). Notably, this association was restricted to CHIP carriers and was not observed in the overall population (odds ratio, 1.03; *P*=0.68) or in noncarriers (OR, 0.99; *P*=0.55). We observed a significant positive interaction between CHIP status and gIL-1RAP on stroke risk, indicating an amplifying effect of gIL-1RAP among CHIP carriers (interaction HR, 1.18 [95% CI, 1.07–1.30]; *P* for interaction=6.06×10^{-4}). A similar pattern was observed for ischemic stroke, with a significant CHIP–IL-1RAP interaction (interaction HR, 1.16 [95% CI, 1.03–1.31]; *P* for interaction=0.014). For hemorrhagic stroke, the direction of effect was consistent, but the interaction did not reach statistical significance (interaction HR, 1.25 [95% CI, 0.99–1.57]; *P* for interaction=0.066). To complement the genetically predicted cytokine analyses with measured protein levels, we examined measured IL-1RAP concentrations in the UK Biobank OLINK cohort (n=51 404). In Cox regression models evaluating the interaction between CHIP and measured IL-1RAP, the interaction was not statistically significant (HR, 1.17 [95% CI, 0.60–2.26]; *P*=0.65), likely reflecting limited statistical power given that only 34 individuals had both CHIP and incident stroke.

DISCUSSION

In this large-scale analysis across 3 independent cohorts, we found that CHIP was significantly

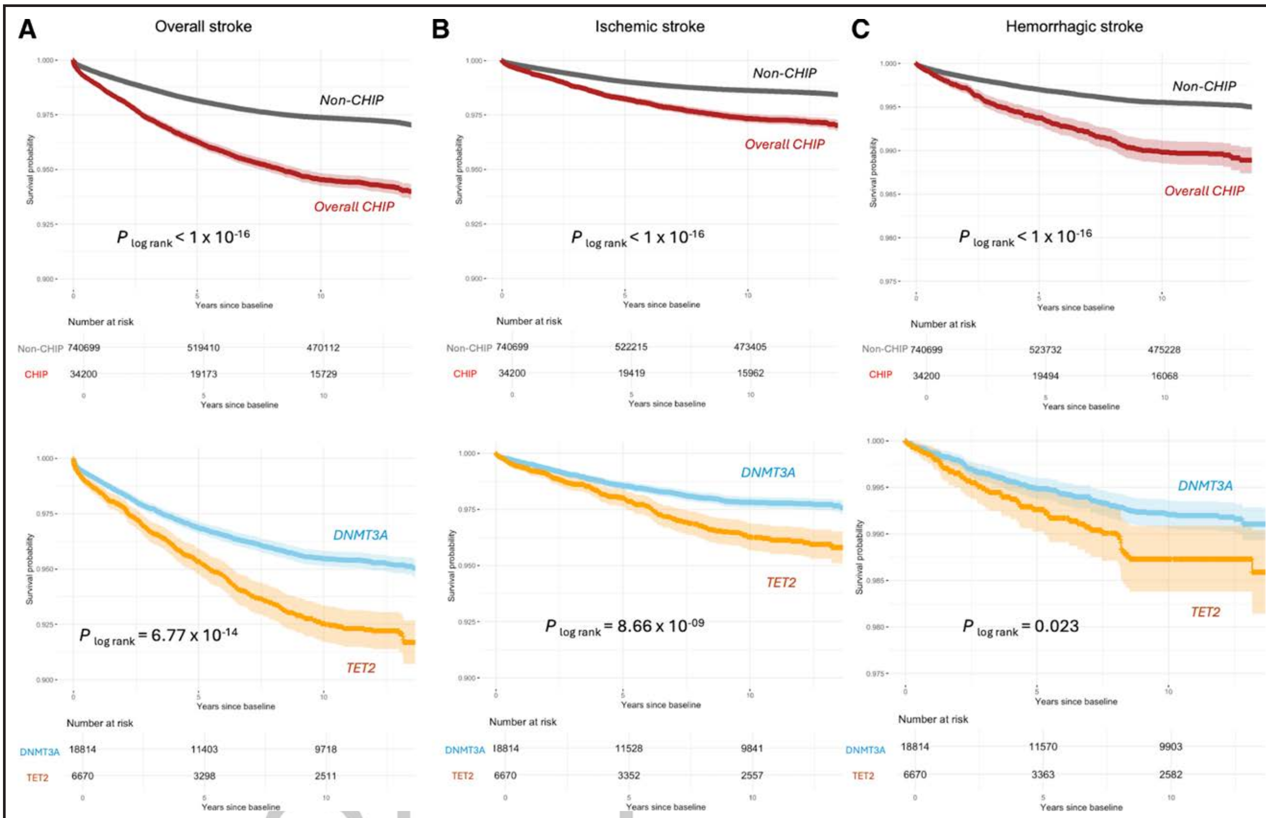

Figure 3. Kaplan-Meier curves for stroke risk according to clonal hematopoiesis of indeterminate potential (CHIP) status and driver genes.

Kaplan-Meier analyses of (**A**) overall stroke, (**B**) ischemic stroke, and (**C**) hemorrhagic stroke in the combined BioVU, All of Us, and UK Biobank cohorts. Participants were categorized into: non-CHIP group (gray line), all CHIP carriers (red line), *DNMT3A*-CHIP carriers (blue line), and *TET2*-CHIP carriers (orange line). CHIP carriers had significantly lower stroke-free survival compared with noncarriers, with particularly steep declines among individuals harboring *TET2* mutations. The number at risk at each time point is shown below the plots.

associated with an increased risk of incident stroke. This association was consistent across cohorts and extended to both ischemic and hemorrhagic subtypes, with relatively higher HRs observed for hemorrhagic stroke. Gene-specific analyses indicated that mutations in *JAK2* and *TET2* were the strongest drivers of stroke risk, whereas *DNMT3A* exerted weaker effects. Analyses stratified by sex and then menopausal status revealed that CHIP was associated with stroke risk in both sexes; however, among women, this association was evident only in those who were postmenopausal. IL-1RAP, a key component of the interleukin-1 signaling pathway, is a potential modifier that enhances the risk of CHIP-related stroke. These findings permit several conclusions.

First, CHIP is a strong risk factor for incident stroke, with an HR comparable to traditional stroke risk factors such as male sex for ischemic stroke and smoking for hemorrhagic stroke. Earlier reports suggested strong associations in smaller cohorts,[24] whereas larger population-based studies yielded more modest or null effects, particularly outside of *JAK2* mutations.[32,33] Interestingly, the HRs were numerically higher for hemorrhagic stroke, which may reflect CHIP-driven inflammatory activation that compromises vascular integrity.[25,39,40] CHIP-associated stroke risk varies substantially by driver gene and stroke type. While it is well-known that *JAK2* CHIP strongly increases the risk of ischemic stroke, the relationship of other driver genes and both types of stroke is more contested. Our results suggest that *TET2* CHIP confers a substantial risk of incident ischemic stroke, but not hemorrhagic stroke. A plausible explanation for the preferential association of *TET2*-CHIP with ischemic stroke is its well-established role in promoting a proinflammatory myeloid phenotype that accelerates atherosclerosis.[29] Experimental studies have shown that *TET2* loss enhances IL-1β–mediated inflammatory signaling and vascular inflammation, processes that are central to ischemic but not hemorrhagic cerebrovascular disease.[41] Although the relative effect size of CHIP on stroke risk is modest, the absolute increase in stroke risk for an individual patient is small. Nevertheless, our findings underscore CHIP as a biologically informative modifier of cerebrovascular risk. At present, CHIP status primarily advances understanding of disease mechanisms linking clonal hematopoiesis, inflammation, and stroke, and

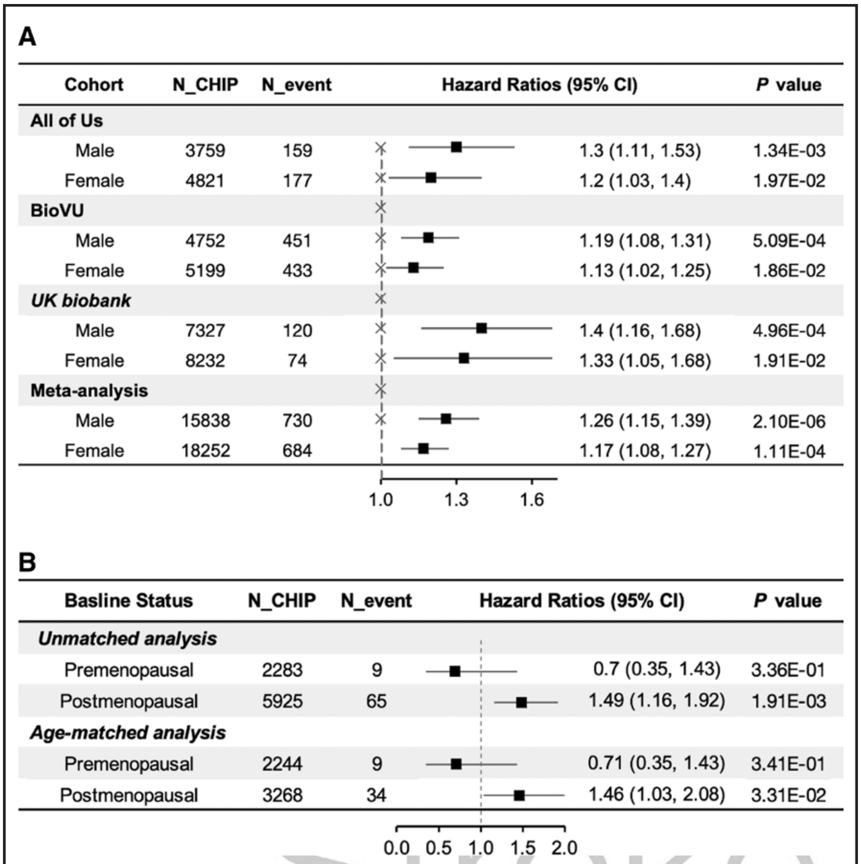

Figure 4. Sex-stratified associations of clonal hematopoiesis of indeterminate potential (CHIP) with incident stroke.
A, Forest plots showing hazard ratios and 95% CIs for the association of CHIP with incident stroke, stratified by sex across the BioVU, All of Us, and UK Biobank cohorts, and combined in meta-analysis. **B**, Hazard ratios and 95% CIs for the association between CHIP and incident stroke are shown separately for premenopausal and postmenopausal women in UKB (UK Biobank). Results are presented for both the unmatched and age-matched analyses. The same Cox regression model was applied in both analyses with the same covariates (including baseline age). The column N_CHIP indicates the number of CHIP carriers included, and N_event represents the number of stroke cases among CHIP carriers.

provides a rationale for future studies evaluating its utility in refined risk stratification and precision prevention strategies.

Second, our subgroup analyses also revealed important insights regarding clone size and hormonal status. The associations of CHIP with stroke were stable across variant allele frequency thresholds (≥2% and ≥10%). These results suggest that the systemic inflammatory signaling present in anyone with CHIP drives increased stroke risk. At the same time, sequencing-related factors may also play a role due to the known limitations of whole-genome sequencing in detecting small clones.[42] For example, most CHIP mutations (72.2%) had VAF ≥10%. Although the CIs of the estimates overlapped, the point estimates suggested that women may have a lower CHIP-attributable risk of stroke than men. When we further hypothesized that estrogen may play a protective role against CHIP-associated stroke risk, we found that in women, the association between CHIP and stroke was evident only among those who were postmenopausal, even after matching for age. However, in premenopausal women, the co-occurrence of CHIP and stroke was rare, given the strong age dependence of both CHIP prevalence and stroke incidence. Consequently, the limited number of events constrained statistical power and precluded definitive conclusions regarding the presence or

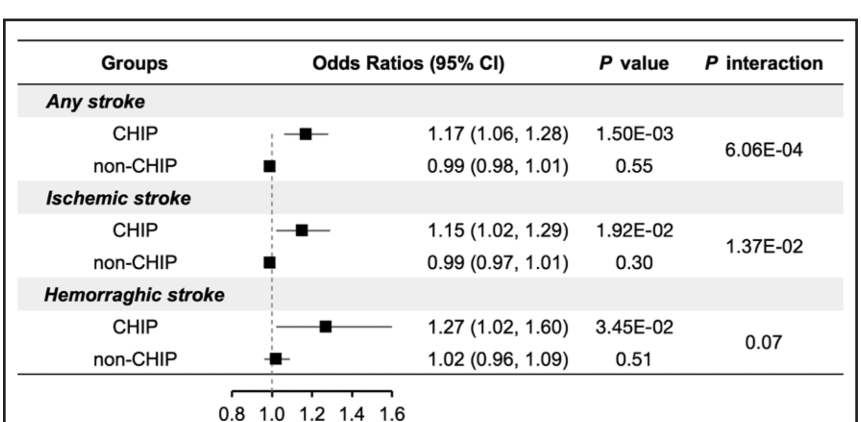

Figure 5. Association of genetically predicted IL (interleukin)-1 receptor accessory protein with stroke risk.
Forest plots display the associations between genetically predicted IL-1RAP (IL-1 receptor accessory protein) levels and risk of stroke among clonal hematopoiesis of indeterminate potential (CHIP) carriers and noncarriers. Odds ratios and 95% CIs were estimated using logistic regression models. Results are derived from meta-analysis across the BioVU, AoU (National Institutes of Health All of Us Research Program), and UKB (UK Biobank) cohorts restricted to participants of European ancestry.

absence of an association. These results suggest that further work should be undertaken to understand how loss of endogenous estrogen, a hormone with known anti-inflammatory, antioxidative, and vasculoprotective effects,[43–45] may amplify the vascular consequences of CHIP-related inflammation.

Third, IL-1 signaling may be a promising drug target for CHIP-associated stroke risk with biologics such as anakinra or canakinumab. CHIP carriers with higher genetically predicted IL-1RAP signaling experience heightened stroke risk. By contrast, this pattern was not observed using measured IL-1RAP levels from the UK Biobank OLINK cohort. This discrepancy may reflect limited statistical power as well as the fact that a single protein measurement is influenced by transient physiological states. IL-1RAP is an essential co-receptor for IL-1 family cytokines, including IL-1, IL-33, and IL-36, and thus amplifies interleukin-driven immune responses.[46] Our human genetic evidence is corroborated by prior work examining the usage of IL-1 blockade in the CANTOS trial (Canakinumab Anti-inflammatory Thrombosis Outcomes) to reduce CHIP-associated disease risk. In those with CHIP, administration of canakinumab demonstrated reductions in major adverse cardiovascular events,[47] inflammatory signaling,[48] and incident solid-organ malignancies[49] exclusively among CHIP carriers. Future work is needed to test prospectively or via analysis of prior samples whether IL-1 blockade may have similar effects for incident stroke.

Strengths and Limitations

Our study has several notable strengths. We analyzed an exceptionally large sample of >800 000 participants, which provided statistical power to detect modest associations and examine rare CHIP mutations. CHIP identification and stroke outcomes were harmonized across cohorts using standardized sequencing pipelines and definitions. We further adjusted for an extensive set of vascular and demographic covariates and performed detailed gene-specific, variant allele frequency, and sex-stratified analyses. Nevertheless, important limitations should be acknowledged. First, stroke ascertainment via *ICD* codes without adjudication (nor primary or secondary classification) reduces the fidelity of hemorrhagic versus ischemic subtype classification and thus limits the conclusions that can be drawn from our subtype-specific associations. Second, despite extensive covariate adjustment, some degree of residual confounding, survival bias, and variation in healthcare utilization may remain. Third, despite harmonized definitions and analytic methods, the 3 cohorts differ in follow-up duration and population characteristics, which may contribute to heterogeneity in stroke event capture and effect estimates; accordingly, meta-analytic estimates with higher I^2 should be interpreted with caution. Fourth, observations regarding IL-1RAP as a potential modifier of CHIP-related stroke risk should be interpreted as hypothesis-generating rather than clinically actionable, and direct biochemical validation will be required. Last, the precision of some subgroup analyses is limited due to the rarity of CHIP within certain strata and the low incidence of stroke events. For example, only 9 strokes occurred among 2283 premenopausal CHIP women, and 8 hemorrhagic strokes among 486 individuals with *JAK2*-CHIP. These results should therefore be interpreted with appropriate caution. These caveats underscore the need for longitudinal sequencing, direct biomarker profiling, and experimental validation to refine our understanding of the mechanisms linking CHIP to stroke.

Conclusions

In conclusion, this study establishes clonal hematopoiesis as an important determinant of incident stroke risk in a genotype-specific manner, with a comparable effect size to traditional risk factors. Our findings suggest that future work should investigate the synergistic effects of CHIP and menopause as women age on stroke and cardiovascular risk. Finally, our human genetic evidence nominates IL-1 as a therapeutic target for the reduction in CHIP-associated stroke risk. These findings underscore the broader relevance of somatic mutations in stroke risk and point toward new opportunities for preventing stroke through genomically informed strategies.

ARTICLE INFORMATION

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Disclosures

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Supplemental Material

Tables S1–S16
Figures S1–S7

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